

App Flow Document

AI-Powered Vinyl Wrapping Visualization System

Screen-by-screen navigation and user journey mapping

Field	Detail
Companion Document	Product Requirements Document (PRD) v1.0
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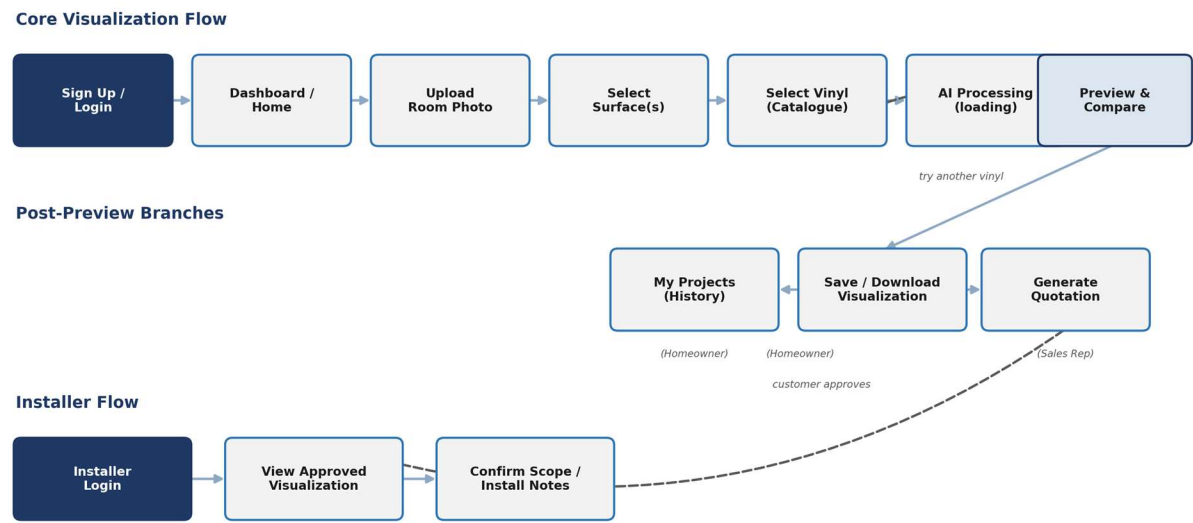
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1. Overview

This App Flow Document maps every screen a user encounters in the AI-Powered Vinyl Wrapping Visualization System, organized by user journey. It complements the PRD by defining navigation paths, screen purpose, on-screen elements, and the actions that move a user from one screen to the next — for all three personas: Homeowner, Sales Representative, and Vinyl Installer.

Flows are read left to right / top to bottom in the diagram below, with dashed arrows indicating optional loops or approval hand-offs between personas.

2. High-Level App Flow Diagram



The core visualization flow (top row) is shared by Homeowners and Sales Representatives. After a preview is generated, the flow branches into saving/downloading, revisiting history, or — for Sales Representatives — generating a quotation. The quotation hands off to a separate Installer flow once the customer approves a design.

Flow 1: Onboarding & Authentication

Entry flow for all personas before reaching their respective home screens.

Screen	Purpose	Key Elements	User Actions	Next
Splash / Landing	Introduce the product and route to sign in or sign up.	Logo, tagline, Sign In / Sign Up buttons	Tap Sign In or Sign Up	Login or Sign Up screen
Sign Up	Create a new account (Homeowner, Sales Rep, or Installer role).	Name, email, password, role selector	Fill form, submit	Email verification / Dashboard
Login	Authenticate an existing user.	Email, password, forgot-password link	Enter credentials, submit	Role-based Dashboard
Forgot Password	Recover account access.	Email input, reset link confirmation	Submit email	Login screen

Flow 2: Core Visualization Flow (Homeowner)

The primary flow: turning a room photo into a photorealistic vinyl preview.

Screen	Purpose	Key Elements	User Actions	Next
Dashboard / Home	Entry point after login; shows recent projects and quick actions.	New Visualization button, recent projects list, catalogue shortcut	Tap New Visualization	Upload Room Photo
Upload Room Photo	Capture or upload the surface/room image.	Camera capture, file upload, image preview	Upload/capture photo, confirm	Select Surface(s)
Select Surface(s)	Mark one or more wrappable components on the uploaded image.	Tappable overlay regions, component list (wall, door, cabinet, etc.), multi-select toggle	Tap/draw surface region(s), confirm selection	Select Vinyl (Catalogue)
Select Vinyl (Catalogue)	Choose a vinyl design to apply to the selected surface(s).	Catalogue grid, filters (color, finish), search	Browse, tap a vinyl to select	AI Processing
AI Processing (loading)	System runs segmentation, texture mapping, and rendering.	Progress indicator, estimated time	Wait (or cancel)	Preview & Compare
Preview & Compare	View the generated visualization; compare against other vinyls or the original.	Before/after toggle, side-by-side compare, try-another-vinyl action	Compare, accept, or try another design	Save/Download, History, or Quotation

Flow 3: Catalogue Browsing

Standalone browsing flow, reachable from the Dashboard without starting a new visualization.

Screen	Purpose	Key Elements	User Actions	Next
Catalogue Home	Browse all available vinyl designs by category.	Category tabs, search, filters	Browse categories, search	Vinyl Detail
Vinyl Detail	View a single vinyl's texture, color code, and metadata at full resolution.	Large texture preview, color code, product name, "Use this design" button	Tap "Use this design"	Upload Room Photo (starts core flow) or back to Catalogue

Flow 4: Saved Projects / Visualization History

Lets users revisit, re-edit, or re-download past visualizations.

Screen	Purpose	Key Elements	User Actions	Next
My Projects (History)	List of previously generated visualizations.	Thumbnail grid, date, surface/vinyl labels, search	Tap a saved project	Project Detail
Project Detail	View a saved visualization; re-edit or re-download.	Full-size image, re-edit button, download button, delete option	Re-edit, download, or delete	Preview & Compare (re-edit) or stays on Project Detail

Flow 5: Sales Representative Flow

Extends the core visualization flow with consultation and quotation tools.

Screen	Purpose	Key Elements	User Actions	Next
Rep Dashboard	Sales-specific home screen with customer session tools.	Start New Consultation button, recent customer sessions	Tap Start New Consultation	Core Visualization Flow (on behalf of customer)
Generate Quotation	Produce a price estimate tied to the approved visualization.	Selected surfaces/vinyl summary, price fields, notes	Fill/confirm pricing, generate	Quotation Summary
Quotation Summary	Shareable summary combining the visualization and price estimate.	Visualization thumbnail, price breakdown, share/export, "Send for approval"	Share with customer or send for approval	Customer approval → Installer Flow

Flow 6: Installer Flow

Read-focused flow for installers to confirm the scope of approved work.

Screen	Purpose	Key Elements	User Actions	Next
Installer Dashboard	List of jobs assigned to the installer.	Job list with customer name, address, status	Tap a job	View Approved Visualization
View Approved Visualization	See the customer-approved preview and selected materials.	Full visualization image, surface list, vinyl ID/color code	Review details	Confirm Scope / Install Notes
Confirm Scope / Install Notes	Acknowledge the job scope and add installation notes.	Checklist of surfaces, notes field, confirm button	Check off items, add notes, confirm	Installer Dashboard (job marked in progress)

Flow 7: Account & Settings

Shared across all personas.

Screen	Purpose	Key Elements	User Actions	Next
Account Settings	Manage profile and preferences.	Name, email, password change, notification toggles	Edit fields, save	Dashboard
Company / Tenant Settings	(Admin only) manage catalogue and team access.	Catalogue upload, team member list, branding	Edit settings, save	Dashboard

10. Full Screen Inventory

#	Screen	Flow	Persona(s)
1	Splash / Landing	Onboarding	All
2	Sign Up	Onboarding	All
3	Login	Onboarding	All
4	Forgot Password	Onboarding	All
5	Dashboard / Home	Core Visualization	Homeowner
6	Upload Room Photo	Core Visualization	Homeowner, Sales Rep
7	Select Surface(s)	Core Visualization	Homeowner, Sales Rep
8	Select Vinyl (Catalogue)	Core Visualization	Homeowner, Sales Rep
9	AI Processing (loading)	Core Visualization	Homeowner, Sales Rep
10	Preview & Compare	Core Visualization	Homeowner, Sales Rep
11	Catalogue Home	Catalogue Browsing	Homeowner, Sales Rep
12	Vinyl Detail	Catalogue Browsing	Homeowner, Sales Rep
13	My Projects (History)	History	Homeowner
14	Project Detail	History	Homeowner

#	Screen	Flow	Persona(s)
15	Rep Dashboard	Sales Rep	Sales Rep
16	Generate Quotation	Sales Rep	Sales Rep
17	Quotation Summary	Sales Rep	Sales Rep
18	Installer Dashboard	Installer	Installer
19	View Approved Visualization	Installer	Installer
20	Confirm Scope / Install Notes	Installer	Installer
21	Account Settings	Account	All
22	Company / Tenant Settings	Account	Admin

11. Navigation Rules & Edge Cases

- Back navigation is always available except during AI Processing (loading), where only Cancel is offered.
- Cancelling AI Processing returns the user to Select Vinyl (Catalogue) with the prior selections retained.
- Preview & Compare supports switching vinyls without re-selecting surfaces — this loops back to Select Vinyl (Catalogue) directly.
- A Sales Rep's consultation session is tied to a customer record; leaving mid-flow auto-saves as a draft in Rep Dashboard.
- Installers only gain access to a job's visualization after the Sales Rep or Homeowner marks the quotation as customer-approved.
- Deleting a saved project from Project Detail requires a confirmation step (not shown as a separate screen).